

Technical Architecture

DELMIA Apriso handles Complex types in a different way than simple scalar data types (for example Char, Integer, Decimal).

Passing Data By Reference

When a simple scalar data type is passed between Functions or DFCs, it is passed by a value. This protects the data from accidental overwriting, because passing by a value means that only a copy of the data is passed.

A Complex variable is passed directly without creating a copy – only a reference to the value is passed. Because a Complex type can store many values, this mechanism makes it possible to manipulate big data objects with the same speed as simple types. But such a mechanism is only possible when data is shared between variables.

The downside of this approach is that when sharing data there is a risk of unintended data modification. To reduce such a risk, DELMIA Apriso allows Process Authors to break the dependency by copying values to a new runtime object.

Simple scalar data types are passed by a value. Complex and all List types are always passed by a reference.

No Direct Conversion From Simple Scalar Types

Because a Complex variable can store any number of values, it is not possible to convert a Complex type to or from any other data type. That is why a dedicated component has been added to Process Builder that allows the user to compose a Complex variable from a number of simple variables and later to decompose it, if needed.

To learn more, refer to [Variables Composition And Decomposition](#).

Nullability

Because Complex types use the "pass by reference" mechanism, it is possible to create a reference without creating the actual value. When a variable does not reference any particular value, we can talk of a `null` value – this can happen both in case of a Complex variable itself and of the Complex variable properties.

The `null` value scenario is not common, but in some situations such a possibility can be an asset, for example when working with database objects.

Trying to access the contents of a `null` value will result in a runtime exception.

Default value of a Complex data type is always `null`.